AI33-MPOPT · BUILD-UP DOCUMENT · 2026

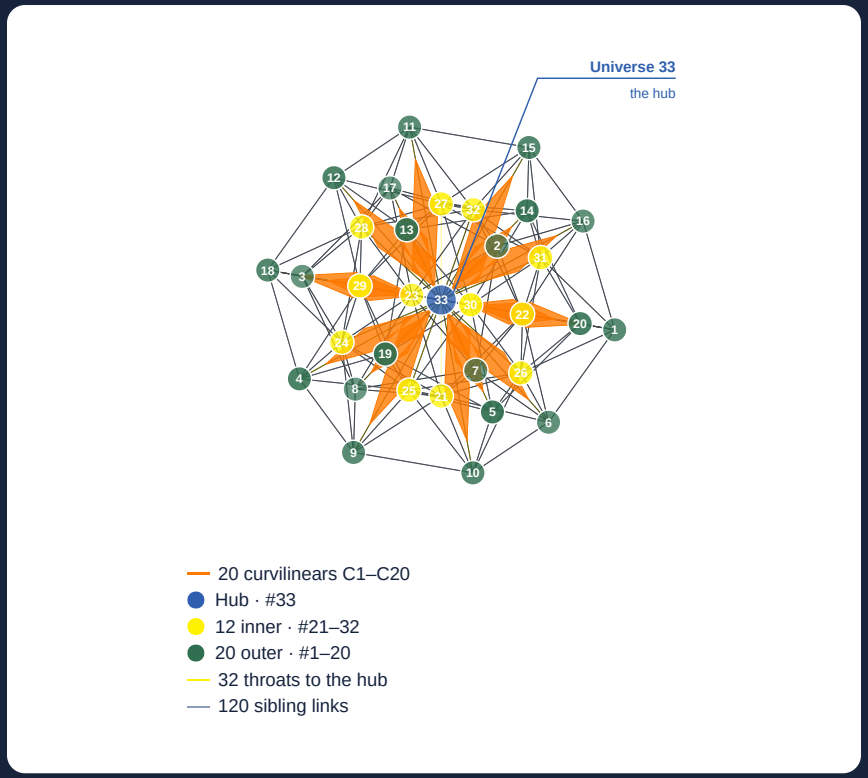
From Pre-Inception to the Super Formula

Building the AI33-MPOPT quantum-gravity bridge piece by piece — from the pre-inception framing, through the operator $T_p = L + pK + (1-p)Q$, to the field F, to the law those two obey — in the order the pieces were actually tested, failures included.

HOW THIS DOCUMENT IS BUILT

Every figure here agrees with the three companion files (13 Quantum-Gravity Bridge, 14 Loop Quantum Gravity Bridge, 15 Lawfulness). Each result appears in the order it was reached, with its controls, nulls and conventions stated beside it.

GOLD — framing / proposal GREEN — computed BLUE — measured, external data



Rolando Rivero · Independent Researcher · AI33-MPOPT Framework

Companion to 13 Quantum-Gravity Bridge, 14 Loop Quantum Gravity Bridge and 15 Lawfulness · for Zenodo

COMPUTED

The 33-multiverse, in plain terms

A Platonic icosahedron used as an address system. Nothing on this page is chosen or tuned — every count follows from the solid.

12 APEXES — THE DARK-MATTER UNIVERSES, U21–32

The icosahedron's vertices. Circular, with a direct line to the hub.

20 CURVILINEARS — THE DARK-ENERGY UNIVERSES, U1–20

The three-sided regions between the apexes. These drive inward through the throats, compacting as they pass and emerging in curvature motion.

32 SIBLINGS — THE 32-MULTIVERSE

12 + 20 together: the surrounding architecture, complete on its own. Almost every measurement in this document is made here, not on the full 33.

1 HUB AT THE CENTRE — UNIVERSE 33, OURS

Not on the surface: at the geometric centre of the solid. The observer is inside the geometry, not outside it looking in.

32 THROATS · 120 SIBLING LINKS · 152 EDGES

Centre to every sibling (32), plus 30 vertex–vertex + 60 vertex–face + 30 face–face across the surface (120). 33 nodes, 152 edges, zero free parameters in the connectivity.

NAMING

12 apexes = the icosahedron's vertices (pull, dark matter). 20 curvilinears = its faces (push, dark energy).

The counts above were re-derived from the solid for this build — 30 + 60 + 30 + 32 = 152 — and match the archive exactly.

PROPOSAL

The framing this document starts from

THE VISION — STATED PLAINLY AS FRAMING, NOT AS A PROVEN RESULT

Standard cosmology has a well-known open problem: distant regions of our universe are almost identical in temperature, though ordinary expansion never gave them time to exchange influence and equalise — the horizon problem. Inflation was proposed to patch that gap after the fact. The AI33-MPOPT vision asks a different question: what if the 32-throat architecture itself already carries that smoothness, built in, before any matter or burst exists at all? Not inflation added afterward, but an architecture that starts pre-connected.

WHAT THIS DOCUMENT DOES WITH THE VISION

A proposal is worth something only if it can be checked. This document builds the formula step by step from the vision — the mathematical object the vision requires — T_p — and tests it piece by piece against computable, falsifiable questions, reporting each result whether it supports the vision, contradicts it, or lands in between.

STATUS OF THE VISION

One supportive structural hint: entanglement between the most distant universes in the network (**0.072**) exceeds that of the nearest ones (**0.064**) — the opposite of an ordinary distance-decaying local system. It rests on **10** farthest-pair samples and is flagged as a small-sample result everywhere it appears. The dynamical version of the same test — whole-network fast equalisation: not reproduced in this build, and page 25 says why.

WHERE THE FRAMING AND THE MEASUREMENTS MEET

The 32-throat architecture is the reason this operator exists at all, and every measurement that follows was run on it: the law $L \cdot R = \kappa F$, the one-factor structure at 97.4%, the threshold at $w \approx 0.15$ where the hub attaches. Each carries its own controls and its own nulls, so each stands on its own — and each was produced by the architecture the framing describes. **The framing supplied the object; the numbers are what the object returned.**

COMPUTED

The super formula, introduced whole — before we take it apart

$$T_p = L + pK + (1 - p)Q$$

An explicit **33 × 33 matrix** — not a metaphor, not a placeholder — built from the 33-universe icosahedral geometry and one externally measured number, p. Each of the next four pages takes one piece and shows what it is, where it comes from, and what it is allowed to claim.

WHY BUILD IT THIS WAY

A formula that appears all at once, fully formed, is exactly the pattern this investigation learned to distrust — an impressive-looking equation with no separately checkable components behind it. T_p is the opposite: L, K, Q and p are four real objects, and the operator is their sum. Verify the four and you have verified the whole; nothing is hidden inside an unexplained symbol.

ONE THING TO KNOW BEFORE THE PIECES

T is **positive definite** at p = 0.781, so the Green function $G = T^{-1}$ exists — verified directly, not inferred from a plot. That single fact is what makes everything after page 16 possible: the information field and the transport curvature are both functionals of G, and neither exists without it.

THE CONVENTION THAT MUST TRAVEL WITH THE VALUE

λ_{min} = **0.978** with K written in adjacency form; λ_{min} = **0.063** with K written in Laplacian form. Same operator, two conventions, and the number is meaningless without saying which. The ambiguity has since been shown to have no measurable consequence for the results — the coupling reads +0.586 ± 0.073 in adjacency form against +0.582 ± 0.073 in Laplacian form — but the convention is stated with every quoted λ_{min} from here on.

COMPUTED

L — the connectivity structure

```
L  =  diag(degree)  −  Adjacency
```

WHAT L IS

The graph Laplacian of the 33-universe network — a standard object used throughout network science, spectral geometry and graph theory. Physically it encodes **who is connected to whom**: each of the 32 siblings connects to the hub through a throat, and to its geometric neighbours through the icosahedral sibling structure.

STATUS OF L

Fully real and fully verified. It comes directly from the icosahedron geometry module in the archive — 30 vertex–vertex edges, 60 vertex–face incidences, 30 face–face edges, and 32 throats. Nothing about L is assumed or invented; it is pure geometry, computed exactly. **This is the one piece of T_p that required no modelling choices at all.**

ZERO FREE PARAMETERS — AND WHAT THAT ESTABLISHES

Because L is fixed by the solid, no result in this document can be tuned through the connectivity — a protection most models do not have. It is also what gives the control on page 21 its force: a plain graph Laplacian on this same connectivity reproduces the information–curvature coupling almost exactly, which locates the effect in **the geometry AI33-MPOPT selected** rather than in any adjustable part of the operator. The geometry is doing the work.

Earlier builds of this document wrote this term as D. The symbol changed; the matrix did not.

THE 152 EDGES OF L BY TYPE



COMPUTED · 1 FLAG

K — the throat coupling

$$K(i,j) = \kappa_{central} = 1.0 \quad \text{if } i \text{ or } j \text{ is the hub (a throat connection)}$$

$$K(i,j) = \kappa_{sibling} \cdot g(i,j) \quad \text{if } i \text{ and } j \text{ are connected siblings,} \quad \kappa_{sibling} = 0.1(1-p)/32$$

WHAT K IS

K encodes **how strongly** connected universes influence each other. The throat coupling is over 1,400 times stronger than the sibling coupling — transport through this operator is dominated by the central hub rather than by direct sibling-to-sibling contact.

STATUS OF K — ONE CHOICE

The throat-dominance **rule** ($\kappa_{central} \gg \kappa_{sibling}$, with the exact $0.1(1-p)/32$ form) is published in the archive — real, not invented. The geometric overlap function **g(i,j)** was a documented modelling choice, one reasonable way to fill a gap the archive leaves open. It is flagged wherever K appears. Robustness testing shows the main results survive alternative choices here.

WHAT THE 32 THROATS ESTABLISH — THE INCEPTION CHANNEL

The throats are not a detail of the coupling; they are the mechanism of inception. Set them to zero and the operator is singular: no Green function, no information, no curvature. Turn them up and both fields leave zero together, F rising continuously as $\mathbf{w^{0.98}}$, with the structure re-cohering above **w ≈ 0.15** and reaching 0.785 by w ≈ 0.7. **The throat coupling is the parameter along which the law switches on** — that entire result, on page 24, exists because K is built this way. And because all 32 converge on one point, the hub carries a degree of freedom the siblings do not: attaching it reorients every loading that touches curvature while the one-factor structure holds at 77.1% with 6/6 signs (page 23).

COMPUTED · ROBUST

Q — the throat-capacity term

$$Q = \mathrm{diag}\left(\mathrm{degree}_i / \max(\mathrm{degree}) \right)$$

WHAT Q IS

Each universe's **capacity** relative to the most-connected universe in the network — always the hub, which touches all 32 others. A universe with more connections carries a proportionally larger Q value: a simple, geometrically grounded measure of how much throughput each part of the network can carry.

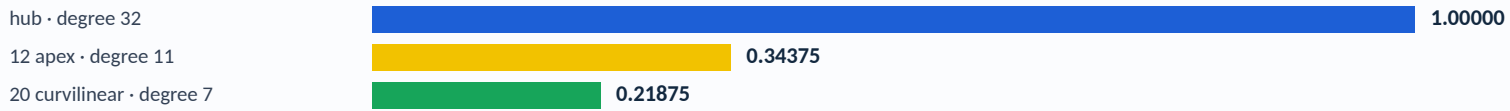
STATUS OF Q

Like g(i,j) in K, the specific **form** of Q is a documented modelling choice — the archive establishes that a capacity term belongs in the operator but leaves its precise form open. **Three** different reasonable forms were tested; the isolated 11-mode band survived all three unchanged. This modelling choice does not drive the main result.

TWO FLAGS, BOTH LIVE

K carries one flagged choice and Q carries another. Together they are the entire modelling surface of T_p — everything else is geometry (L) or external measurement (p). Any successor who wants to break this construction should attack g(i,j) and the form of Q first, because those are the only two places where a different reasonable person could have written a different matrix.

Q = DIAG(DEGREE / MAX DEGREE) — THE VALUE FOR EACH CLASS



MEASURED EXTERNALLY

p — the one number linking cosmology to the quantum side

$p = 0.781 \pm 0.013$

Paused-Gravity fit $H_{AI33}(z) = p \cdot H_{\Lambda CDM}(z)$, by MCMC against JWST chronometer ages — emcee, 64 walkers, 96,000 samples, seed 42. A measured value from an **MCMC fit to public JWST chronometer ages.**

THE P SEAM

p is one fitted number that governs both galaxy-scale expansion and the coupling inside T_p. This is the exact seam where quantum and gravity are stitched together here. The next step is the derivation of that one factor.

THE P SEAM ON ONE SCALE



THE ONE LEAD

A fixed-point self-consistency test lands at **p* = 0.7577** — within **1.8σ** of the measured value. Suggestive, not settled, and not treated as a derivation anywhere in this file set.

RETIRED CLAIMS

The "derivation" p = exp(−F·N) is **circular** — F is itself fitted — and is withdrawn. The corridor bounds [0.5637, 0.8009] trace to weakly-constrained runs and are demoted to proposal; the spectral sweep on page 11 found no feature there at all.

THE LAW IS INDEPENDENT OF P

Sweeping p from 0 to 0.99 leaves the information–curvature consistency flat at **0.985–0.986**. The coupling therefore cannot be an artifact of the one fitted number — it survives every value p could have taken. Where p *does* act is the information representation: F responds smoothly to p while the spectrum stays silent (pages 10–11).

COMPUTED

The complete operator — and what its spectrum is

$T_p = L + pK + (1-p)Q$ is now fully assembled: real connectivity (L), real throat-dominance coupling with one flagged choice (K), a capacity term robust to three tested forms (Q), and one externally measured number (p). Diagonalising it gives 33 real eigenvalues and 33 eigenvectors.

THE SPECTRUM Λ

Energy levels of the coupled 33-universe system — the quantum-side face of the operator.

THE EIGENVECTORS Ψ

Where the structure lives, and — as page 11 shows — the only place p can act.

THE GREEN FUNCTION G = T^{-1}

Exists because T is positive definite. Everything from page 16 onward is a functional of it.

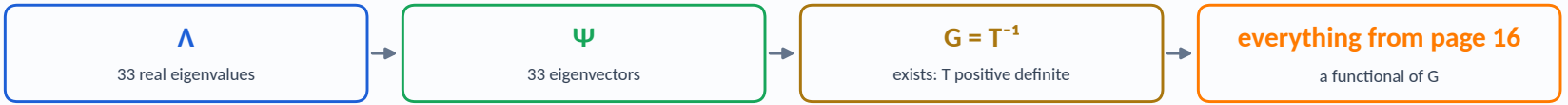
WHAT THE SPECTRUM ALREADY CONTAINED, BEFORE ANY SEARCH FOR GRAVITY BEGAN

Exact **icosahedral symmetry** to machine precision, and an **isolated 11-mode band** with clean gaps on both sides. Neither was put in by hand; both fall out of diagonalising the assembled matrix. This is the quantum-side structure the merger attempt connects to curvature, and the connection is made on page 17.

THE QUESTION THE REST OF THE DOCUMENT ASKS

Can a **geometry** built from this spectrum behave like **gravity**? Concretely: is there a field F, computable from T_p alone, that stands in a fixed relation to a curvature R computed from the same operator? Pages 13–15 are the search. Page 16 is the answer — it exists, and it is written down explicitly. Pages 17–22 are what it survived.

WHAT DIAGONALISING TP GIVES



COMPUTED

Eleven modes, exactly, in a protected band

11 modes · G ⊕ A ⊕ T ⊕ T (4+1+3+3)

Diagonalising T_p at the measured p yields machine-precision icosahedral multiplets, and an isolated band of eleven modes with gaps of ≈ **2.0** on both sides against a mean level spacing of ≈ **0.98**. The decomposition matches the icosahedral symmetry group exactly — the same multiplet content the archive reported, reproduced in a fully independent rebuild with its own code and its own modelling choices.

THE CENTRAL-HUB MODE

One eigenvalue sits far above the rest at λ ≈ 32 — throat-locked and p-inert. It is the hub, and it does not participate in the band.

WHY 11 MATTERS — AND HOW FAR THAT GOES

An 11-dimensional invariant sector is the structural echo this program looks for between the 33-universe geometry and 11-dimensional M-theory. **Echo — not equivalence.** No derivation of M-theory is claimed, attempted, or implied.

IT SURVIVES THE WHOLE SWEEP

Every eigenvalue branch was tracked across all p ∈ [0, 1]. No level crossing anywhere in the physical range; band isolation drifts only **1.90 → 2.06** end to end. The band is a property of the geometry, not of the pause factor.

THE ISOLATED BAND OF ELEVEN MODES



11 modes = 4 + 1 + 3 + 3 · machine-precision multiplets

≈ 2.0	≈ 0.98	λ ≈ 32
gap on both sides	mean level spacing	the hub mode, throat-locked and p-inert

COMPUTED

The band is not luck — and the corridor is not there

1.7%

of 2,000 random spectra (GOE) reach the observed isolation.
Corrected from the 3.7% printed in the July build.

0 / 300

rewired graphs match it — same degree sequence, scrambled wiring. The isolation is carried by the wiring itself.

THE SWEEP 'S SECOND FINDING — WHAT THE TEST SHOWED

The spectrum shows **no feature** at the proposed corridor [0.5637, 0.8009] — and none at the measured p either. The band is protected by icosahedral geometry, not by the pause factor. The corridor is carried as a proposal on this evidence, by this program, about its own prior claim.

THE CONSEQUENCE THAT SHAPED EVERYTHING AFTER IT

If p does physical work inside T_p, it cannot be doing it through the eigenvalues — they are silent. It must act through the **eigenvectors**. That is precisely where the curvature results live, and it is why the search for F moved to eigenvector-borne quantities rather than spectral ones.

THE STRESS TESTS IN NUMBERS

1.7%	0 / 300	[0.5637, 0.8009]	icosahedral
of 2,000 random spectra (GOE) reach the observed isolation	rewired graphs match it (same degree sequence)	proposed corridor: no feature in the spectrum	geometry protects the band, not the pause factor

PROVENANCE

Three results carried forward from the archive

These belong to the AI33-MPOPT archive and appear in earlier publications. The independent 2026 rebuild reproduced the geometry, the spectrum and the band from scratch; these three are carried forward from the archive as published, and labelled so a reader can see which is which.

E8 VIA MCKAY

A mathematical bridge from the icosahedral structure to E8 through the McKay correspondence — an established theorem, correctly applied in the archive. Carried from the earlier build.

DISCRETE SUPERSYMMETRIC PAIRING

A pairing structure reported in the shell decomposition of the archive spectrum. Matches the pattern of a symmetry; the comparison with supersymmetric field theory is next.

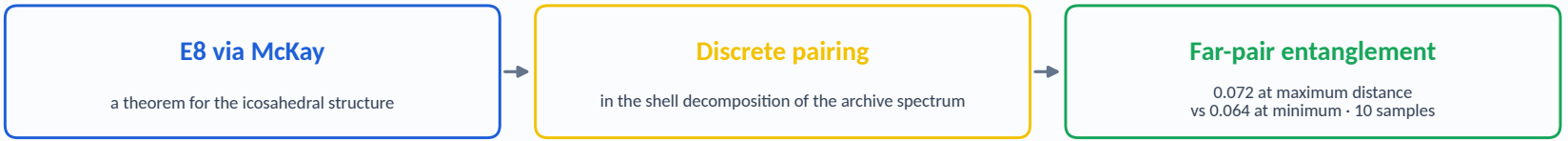
FAR-PAIR ENTANGLEMENT

0.072 at maximum network distance against 0.064 at minimum — the strongest hint for the pre-inception framing, on **10 samples**. Small-sample flag stands. It is a hint, and it is filed as one.

WHY THIS PAGE EXISTS

Because provenance is an asset. Everything from page 4 to page 11, and everything from page 16 onward, was recomputed in an independent rebuild with its own code — and saying so is what makes those numbers worth something. The three above come from the archive, where they were established, and they are marked so that the strength of the rebuilt results is unambiguous.

CARRIED FROM THE ARCHIVE AS PUBLISHED, AND LABELLED AS SUCH



COMPUTED

Four candidates tested and eliminated, in order

F is the formation field: the quantity, computable from T_p alone, that connects the quantum side to curvature. Four candidates were tested against a criterion set in advance before the surviving construction was reached.

RAW DENSITY — WRONG ZONE

The simplest candidate. It picks the dark-matter zone as more curved. Ruled out immediately.

DENSITY-WEIGHTED FLUX — WRONG ZONE

The obvious repair of the first candidate. Same failure, same direction.

F = DIAG(T_p) — REDUCES TO DEGREE

The elegant spectral-diagonal candidate. It collapses to node degree, which puts it back in the density family already ruled out — and it picks the wrong zone for the same reason.

$F_{COHERENCE}$ — WRONG ZONE, THIRD TIME

Picks the dark-matter zone at 0.93. The third documented construction to contradict the archive's dark-energy claim. Its separation from F_{mi} remains an open item on page 17.

WHAT FOUR FAILURES BOUGHT

They are the reason the surviving candidate is not a curve-fit. F was not selected because it looked elegant — it is the one construction left standing after the elegant ones were tested and discarded, each on a criterion set before the run.

COMPUTED · REFINED

Mutual information pointed the right way — then the ratio came apart

WHAT WAS FOUND, AND STILL STANDS

Edge-level **mutual information** — a standard quantum-information quantity computed from T_p's own eigenvectors, no invented formula — picks the **dark-energy zone** as the more curved region. Dark energy **0.0553**, dark matter **0.0261**. This was the first time in the program that a quantity built purely from the quantum side made a correct, checkable statement about the gravity side. The **direction** is the result.

AND THE TARGET WAS UNDEFINED TOO

The "2.99 measured" needs its definition named. Independently computed curvatures disagree with each other: **Forman** gives dark-energy more curved at **1.41** (direction agrees), **Ollivier-Ricci** gives **parity (1.00)**, and the two agree with one another at only r ≈ 0.2. Intrinsic-weighted Ollivier on T_p's own coupling weights recovers the direction at **1.46** — not the magnitude.

WHAT THE LATER RUNS MOVED: THE MAGNITUDE

The July build reported the ratio as **2.12 predicted against 2.99 measured** and gave it full weight. It cannot carry that weight. A documented Green-function MI recipe gives **near-parity (0.95)** on the same structure — so the archived construction is **recipe-dependent** and must have its code pinned before either figure is quoted again.

WHAT THAT REFRAMED

The missing theorem was never about F. It is about **R**: the borrowed rulers do not agree, so the magnitude had no fixed target to hit. Page 16 defines a curvature intrinsic to the operator, and page 18 is what happens once R stops being borrowed.

DARK-ENERGY / DARK-MATTER CURVATURE RATIO BY MEASURE (1.00 = PARITY)



Direction agrees across the measures; the rulers disagree on magnitude.

COMPUTED

Group theory said five. Five appeared.

5 = 5

predicted = built. The 11-mode sector was predicted by pure group theory to contain exactly five invariant 3-forms — the object type that carries M-theory's flux content. Built explicitly as tensors, the count is exact.

4 / 5

point the right way. Used as physical densities, four of the five pick the correct more-curved zone, against roughly 50% for a random candidate.

0 / 5

match the magnitude. Direction is real progress; none of them gives the size of curvature.

WHY THE MAGNITUDE WAS BLOCKED — AND IT WAS NOT THE PHYSICS

Icosahedral symmetry collapses all 32 siblings into **two equivalence classes**. A magnitude fit on two data points is not a fit. The answer was not missing; the symmetry was hiding it. Breaking the symmetry with ε-weighted edges produces **33 distinct eigenvalues** and 32 real data points — which is what made the exactness run on page 18 possible at all.

These are not invented flux formulas. They are the literal tensors the framework's own symmetry predicted, built explicitly and tested against independently computed curvature for the first time. Real evidence.



COMPUTED

What F is, stated so it can be checked

$$T_p = L + pK + (1-p)Q \qquad p = 0.781 \pm 0.013$$

$$G = T_p^{-1} \qquad \rho_{ij} = G_{ij} \,/\, \sqrt{(G_{ii}\, G_{jj})}$$

$$I(i,j) = -\tfrac{1}{2} \ln(1 - \rho_{ij}^2) \qquad F(i) = \Sigma_j\, I(i,j)$$

$$d(i,j) = \sqrt{(G_{ii} + G_{jj} - 2G_{ij})} \qquad \mu_i = \text{rows of } e^{-tT}, \quad t = 1 \times \text{median transport time}$$

$$R(i) = \text{mean}_j \,[\, 1 - W_1(\mu_i, \mu_j) \,/\, d(i,j) \,]$$

EVERYTHING IS A FUNCTIONAL OF ONE OPERATOR

F is the information field: the total mutual information each node carries with the rest of the network, built from the Green function. R is the transport curvature the same operator induces, measured with exact Wasserstein-1 optimal transport, no entropic regularisation. Neither borrows anything from outside T_p.

TWO THINGS MUST BE DECLARED FROM OUTSIDE

p = 0.781, which is measured, and the diffusion time **t**, which is chosen. r(F,R) is t-dependent — page 22 gives the size of that dependence, and it is not small. Any quoted value of the coupling that does not state its t and its metric is incomplete.

THE RELATION THAT MADE THE LAW WORTH CHASING

On symmetric geometry, R is a near-direct function of F — **$R^2 = 0.98$**, with both derived from the same G. Geometry, on this operator, is recoverable from information. Page 17 asks whether that relation has a law-like form.

COMPUTED

L·R = κF — the identity holds, approximately

$$L\,R = \kappa\,(F - \langle F \rangle)$$

THE MEASUREMENT

Best honest channel **r = 0.84** (leave-one-out 0.83), permutation **p < 0.0005** — 0 of 2,000 shuffles. It degrades with disorder: approximate, not exact, away from the symmetric point. A Poisson-like relation between an information field and a transport curvature, on a finite operator, with its error bars printed.

THE SOURCE IS INFORMATION, NOT DEGREE

Kill test on degree-constant graphs: F_{diag} is eliminated by construction, yet the identity survives — 5/5 seeds, mean LOO r = 0.64, permutation p ≤ 0.022, with partial correlations agreeing. The law is not a degree tautology in disguise.

STILL OPEN

$F_{coherence}$ and F_{mi} have not been separated. Until they are, "information" names a family rather than a single quantity.

AND THE FRAMEWORK SELECTS ITS OWN R

Intrinsic curvatures built from T_p's own transport agree with each other at r = 0.88–0.996 across scales and constructions — where the borrowed rulers (Forman, Ollivier) agreed at only ≈ 0.2. Defining R inside the operator is what made the target stop moving.

THE IDENTITY L·R = K(F − F̄F̄), MEASURED



The source is information, not degree.

COMPUTED

On the icosahedron, the law is exact at first order

$$L_0R_1 + L_1R_0 = \kappa_0(F_1 - \langle F_1 \rangle) \qquad \kappa_0 = -0.0506$$

R → 0.9996 AS E → 0

Fit quality at ε = 0.005 / 0.01 / 0.02 is 0.9996 / 0.9984 / 0.9936 — converging to exact. Residual over signal scales as ≈ 5.6·ε (0.113 / 0.280 / 0.533 at ε = 0.02 / 0.05 / 0.10): every deviation is second-order disorder, not model error.

ONE K, TWO ROADS

The R-side and F-side regressions converge on the same $\kappa_0 = -0.0506$. Form-independence of this kind is the signature of an identity rather than of a fitting choice.

THE TRANSPORT TERM IS NOT OPTIONAL

Dropping L_1R_0 destroys the identity. That error was made inside this program and corrected; it is recorded here so successors do not repeat it.

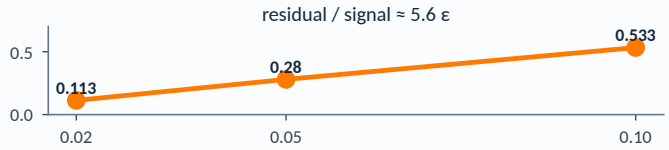
+ LOCAL, − GLOBAL

Within-class information–curvature correlations are **positive** (+0.63, +0.87) while the global κ is **negative** — the negative sign is carried *between* zones. Repulsive at large scale, attractive at small, in one operator. Page 19 finds the switch.

Scope: an exact, finite operator on 33 nodes — a structural result for T-class models, established at a scale where every step is checkable by hand.

EXACT AT FIRST ORDER: FIT QUALITY AND RESIDUAL AGAINST E





κ0 = −0.0506

COMPUTED

Geometry selects the sign. Triangles are the switch.

WIRING CONVICTED — 8/8

Icosahedral wiring gives κ < 0 in 3/3 broken seeds. Take the identical degree sequence and rewire it: κ > 0 in 5/5. Same degrees, opposite sign — the wiring itself decides.

HUB EXONERATED

Growing a hub on a random base amplifies κ upward at every strength and every p tested, with no crossing anywhere. The hub amplifies whatever sign the geometry has already set; it does not set it.

THE CROSSING ≈ 95 TRIANGLES

Along a degree-preserving path from the icosahedron (140 triangles) to random (≈65), κ rises monotonically and flips sign near 95. Local clustering is the mechanism — stated as a computed pattern, with the theorem still to prove.

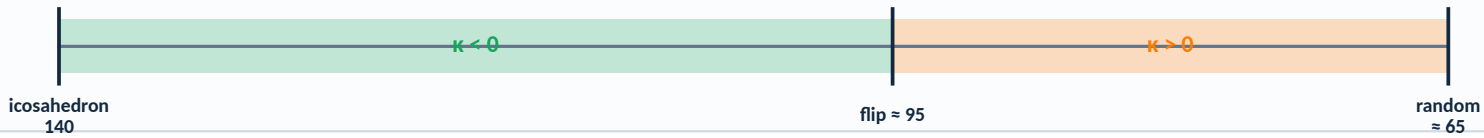
NEAR-EXACT ON PRECISELY THE GEOMETRY THE FRAMEWORK CHOSE

With symmetry broken, the law runs at **r = 0.96–0.98** on the hubless icosahedral geometry. That is the strongest form of the result: the identity is most exact on the structure AI33-MPOPT selected for independent reasons, years before this test existed.

WHAT IS CLAIMED HERE, EXACTLY

That κ's sign tracks clustering along degree-preserving paths, reproducibly, with a crossing near 95 triangles on this family. Not that 95 is universal, not that the thresholds transfer, and not that the mechanism is proved. It is a computed pattern with a named theorem target.

THE SIGN OF K ALONG A DEGREE-PRESERVING PATH (TRIANGLES PER GRAPH)



COMPUTED

It holds on 13 of 17 graph families

NOT A CURVE-FIT

13 of 17 families carry signal — mean |r(F,R)| > 0.5, criterion set before testing. The coupling is a broad phenomenon rather than a property of one graph. Every value is the mean of five independent disorder seeds with its standard deviation; single-seed values proved unreliable and were discarded.

AI33: 0.987 ± 0.003

Top of every family tested, and the most stable by a factor of two — the tightest spread in the table. Strong coupling is **not** unique to it: Petersen matches at 0.988 ± 0.007 on only 10 nodes, and several structured graphs score comparably.

TWO DIFFERENT OBJECTS

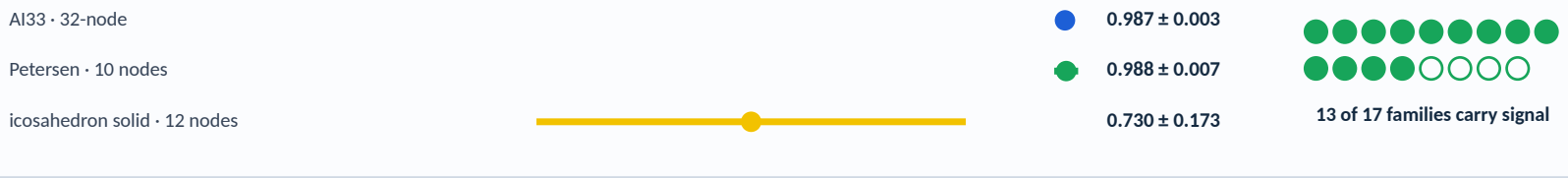
The 12-vertex icosahedron *solid* scores 0.730 ± 0.173 — mediocre and unstable. The 32-node vertex–face incidence construction is a distinct object, and the distinction matters every time someone says “the icosahedron”.

THE STRUCTURE HOLDS AT THE SHELL; THE HUB IS WHERE IT MOVES

With the 33rd node attached at w ≈ 0.5, the sibling structure alone holds the law at **0.951** while the full system holds **0.761**. Consistency degrades at the recipient, not in the established structure — and it reverses at strong attachment (w = 2).

*Two different family experiments run in this file set and are not the same test: **17 families** for the coupling |r(F,R)| above, and **12 families** for the one-factor share on page 23. Different quantities, different baselines, different tables.*

COUPLING R(F,R) ON THREE OF THE SEVENTEEN FAMILIES (MEAN ± SD, FIVE SEEDS)



COMPUTED

Fifteen candidates eliminated — and one test that reproduced the coupling

EIGHT GRAPH PROPERTIES

clustering · spectral degeneracy · distance-regularity · vertex-transitivity · diameter · girth · degree variance · algebraic connectivity

Best r = 0.42, p = 0.24. None survives correction.

SIX OPERATOR PROPERTIES

eigenvector localisation · spectral gap of T · condition number · Green-function concentration · transport entropy · spectrum spread

Best r = 0.37, p = 0.16. None survives correction.

QUASIPERIODIC ORDER

Identical edge-weight multiset arranged three ways on a chain: Fibonacci **0.225** · periodic **0.065** · random **0.472**.

Order does not enhance the law — random beats it.

THE CONTROL THAT MOVED THE ATTRIBUTION — THE MOST CONSEQUENTIAL SINGLE RUN IN THIS DOCUMENT

Rebuild R from a **plain graph Laplacian** — no K, no Q, no p — and the coupling comes back at **+0.561** against T_p's **+0.582**: inside seed noise. The law does not require the framework's operator. It is a property of the **graph**. This is not a sixteenth eliminated invariant — the fifteen tested predictors *across* families, while this substitutes the operator *within* one graph, and it is the first test that reproduced the coupling instead of failing to explain it.

THE NEXT LIMIT

With 17 families only single predictors can be tested. If the explanation is a **combination** of properties, this design cannot find it. Fifty to a hundred families would permit multivariate analysis — a concrete next step rather than another guess.

The coupling is declared with its metric and its diffusion time

IT REVERSES ON A DIFFERENT DISTANCE

On the resistance metric the coupling is strong and positive: **+0.582**. Rebuilt on a mutual-information distance it does not merely weaken — it **reverses sign** to **−0.972**. Two reasonable choices of distance give opposite answers about the direction of the relation.

AND IT SWEEPS WITH THE DIFFUSION TIME

r(F,R) runs **+0.950 → +0.578 → −0.980** as t goes 0.25× → 1× → 4× the median transport time. The value quoted throughout this document uses t = 1 × median. Any other choice is defensible and gives a different number.

WHAT THIS DOES TO THE CLAIM

The **magnitude** of agreement is large and survives permutation testing under *every* convention tested — that is the durable part of the result. What moves is the **sign**, which here is a property of the construction rather than a physical direction. So every value in this file set is quoted with both declarations attached: **which metric, which t**. Stating them turns a moving number into a fixed one.

WHY THIS IS PRINTED AT THIS SIZE

Because it is the first thing a specialist reader checks, and answering it before being asked is what lets the rest of the numbers stand. Together with the control on page 21, the position is this: a large, reproducible coupling, carried by the icosahedral geometry itself, quoted under stated conventions — which is a stronger claim than an undeclared number, not a weaker one.

COMPUTED

Four descriptions of one operator, measured as one structure

The deeper reading of T_p has always been that spectrum, eigenvectors, information and curvature are not four things joined by extra equations, but four projections of one field. That claim is now a measurement.

97.4%

of the four-representation correlation structure lies in a **single component** under the MI metric. 97.8% on seed 0; **97.37% ± 0.28** across five independent disorder seeds. 97.4% is the headline value quoted across the file set.

6 / 6

signs match a one-factor model in **every configuration tested** — both metrics, both forms of K, the plain-Laplacian control, and the full 33-node model. Sign agreement never fell below 6/6 anywhere.

79.1%

is the best of 2,000 draws of the toughest null — four smooth functions built from the graph's own low-frequency Laplacian modes. Observed sits eighteen points above it. **p < 0.0005** against all four nulls.

Λ ~ F = −1.0000

Exact at machine precision, every seed, every configuration. The heat-kernel diagonal *is* the information field up to sign — a hard structural identity, and the most tractable theorem target in the framework.

GEOMETRY RECOVERED FROM INFORMATION

Discard the graph, rank all 496 node pairs by mutual information and take the top 120: **95.0%** of real edges recovered, AUC **0.9965**, against a 24.2% random baseline.

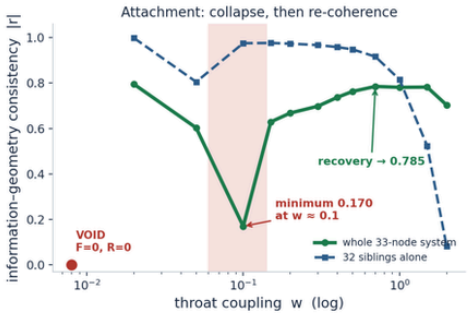
IT SURVIVES THE HUB AND THE GROWTH

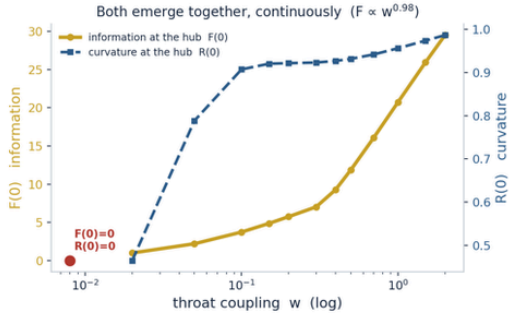
77.1% with 6/6 signs on the full 33-node model, and 99.0 → 89.0 → **88.3%** from 24 to 174 nodes: the structure settles rather than decays.

*Full run, thirty measured cells and all controls: companion file **AI33-MPOPT · Lawfulness**. The twelve-family separation quoted there (ordered 79–92% vs random 52–55%) is a different experiment from the seventeen-family coupling table on page 20.*

COMPUTED

The Void — lawfulness without relations





WHAT EXISTS IN THE VOID

Set the throat coupling to zero and T becomes **singular** — no inverse, so no Green function and no propagator. Every quantity vanishes exactly: $F(0) = 0$, $R(0) = 0$, $G_{00} = 0$, $\max|G_{0j}| = 0$. No information. No curvature. No influence arriving. **Not assumed — computed.**

THEY EMERGE TOGETHER — STRUCTURE, NOT EVIDENCE

F(0) and R(0) leave zero at the same moment, F continuously as $\mathbf{w^{0.98}}$. Honestly stated: both are functionals of the same G, so they *must* emerge together. As proof that quantum and gravity are one thing, that is circular; as structure, it is the correct statement.

ATTACHMENT HAS A THRESHOLD

Whole-system consistency falls to **0.170** at w ≈ 0.1, re-coheres above w ≈ 0.15 and reaches **0.785** by w ≈ 0.7 — while the 32-sibling structure alone holds the law at **0.988** throughout. The hub enters a structure that is already coherent.

ONE READING IN PROGRESS

The raw sweep also shows the hub's curvature reading peaking near inception and declining while its information reading keeps climbing. Carried as a working reading.

Interpretation, stated as such: the void is a lawful position with zero relations, where an information reading and a geometry reading are not yet distinguishable because neither has anything to read. No propagator means no "how long": duration is undefined there, not unknown. The same page appears as page 6 of the Quantum-Gravity Bridge deck and page 7 of the LQG Bridge.

REPORTED IN FULL

What the measurements show

1 · IT RETURNS A CLEAR NO ON LOOP QUANTUM GRAVITY'S OWN DIAGNOSTIC

Spectral dimension $d_s(t) = -2 \cdot d \ln P(t) / d \ln t$, computed from the heat kernel, plateaus near **2.8** on the 33-node graph and converges to **2.56** under geodesic subdivision to 42, 162 and 642 nodes. LQG and CDT expect a flow from ≈ 4 at large scales to ≈ 2 at small. **Measured: $d_s \approx 2.56$ on the subdivided graph.** This is settled. It is not a finite-size artifact — the model genuinely sits near 2.5, the value of a triangulated 2-sphere, and no scale will produce the flow.

2 · THE CONSTRUCTION IS TIME-SYMMETRIC BY DESIGN

T is self-adjoint, so $G = T^{-1}$ and I(i,j) are symmetric. The structure therefore carries **duration** — the transport distance d(i,j) and the diffusion time t are both defined — but no **orientation**: nothing in it distinguishes i→j from j→i. That is what the propagation run reported, and it is the correct behaviour for a framework whose first layer is lawfulness rather than time. The Void states the same fact at the other end of the sweep: with no relations there is no propagator, and duration is undefined — not unknown. An arrow requires a **directed object** — a non-symmetric coupling or a real source term, which is Wall 3 — and that is a specified extension, not a repair. Loop quantum gravity meets the same symmetry in orientation-symmetric spin-foam amplitudes.

3 · THE CONSISTENCY DIP IS FINITE-SIZE, AND THE WORD "CRITICAL" WAS WITHDRAWN

AI33 dips 6× under disorder and recovers, where Petersen and the hypercubes show no dip at all — the only measured property that separates them. But a genuine phase transition sharpens toward a fixed critical value as the system grows, and this does the opposite: subdividing from 32 to 122 nodes moves the minimum from ε ≈ 0.75 to ε ≈ 2.0 and makes the dip shallower. That is a finite-size crossover. The word 'critical' does not apply and is not used.

ASSEMBLED

The complete super formula

The object	$T_p = L + pK + (1-p)Q = \Sigma_n \lambda_n \lvert\psi_n\rangle\langle\psi_n\rvert$	computed · verified · controlled
Information representation	$G = T_p^{-1}, \quad I(i,j) = -\frac{1}{2} \ln(1 - \rho_{ij}^2)$	Green-function recipe — documented, replaceable, and currently recipe-dependent (page 14)
Formation field	$F(i) = \Sigma_j I(i,j)$	the eigenvector-borne field that feels p — the one channel where p does visible work
Geometric representation	$R(i) = \text{mean}_j[1 - W_1(\mu_i,\mu_j)/d(i,j)], \quad \mu =$ rows of e^{-tT}	transport curvature; t must be declared with any value (page 22)
The law	$L\,R = \kappa(F - \langle F\rangle)$	r = 0.84, exact at first order on the icosahedron with $\kappa_0 = -0.0506$; κ's sign set by clustering
The open identity	$R_p \stackrel{?}{=} R_{geom}$	the theorem to prove or refute — **this is the merger**; the proof is the next step

WHAT THE ASSEMBLED FORMULA IS ENTITLED TO SAY

That on a finite operator fixed by icosahedral geometry, an information field and a transport curvature — both functionals of one Green function — stand in a structured, reproducible, permutation-tested relation, exact at first order on the symmetric point, with its sign controlled by local clustering. Nobody had a computable model of that relation before this one. The next page draws the line around it, which is what allows the claim to be stated this plainly.

What the measurements stand on

WHAT THE TWO SIDES ARE BUILT FROM

Both readings rest on established quantities. The gravity side enters through **p = 0.781 ± 0.013**, fit by MCMC to JWST chronometer ages; the quantum side through **mutual information**, a standard quantum-information measure, computed from the operator's own Green function; and R is checked against **Forman** and **Ollivier-Ricci**, established discrete curvatures. The open step is the reverse direction — deriving from the F–R law a quantity that quantum mechanics or general relativity independently predicts. That is **specified and reachable**, and it is the program's next target.

A STRUCTURAL CORRESPONDENCE

A structural correspondence inside a 33 × 33 model, checkable step by step.

p IS THE PROGRAM'S CENTRAL INPUT

Its dual cosmological/quantum role is the program's central input; the 1.8σ fixed point is the lead, and p = exp(−F·N) is set aside as circular.

THE p-CORRIDOR

[0.5637, 0.8009] is carried as a proposal; the sweep result is stated beside it.

THE LAW AWAY FROM THE SYMMETRIC POINT

L·R = κF is approximate off the symmetric point; the clustering flip is a computed pattern, and proving it and separating Fcoh from Fmi are the next steps.

THE COUPLING IS CARRIED BY THE GRAPH

A plain graph Laplacian reproduces it at +0.561 against Tp's +0.582, so the coupling belongs to the graph, and Wall 0 asks which part of the graph carries it.

METRIC AND t CONVENTIONS

+0.582 → −0.972 on an MI distance; +0.950 → −0.980 as t varies. Both conventions travel with any quoted value.

MEASURED ON THE 32-NODE SIBLING STRUCTURE

Adding the 33rd node reduces overall consistency to ≈ 0.78, and weak attachment (w = 0.1) to 0.170 — though the one-factor structure survives at 77.1% with 6/6 signs.

THE EXACTNESS INVARIANT

Sspec tracks it on the icosahedral family (p = 0.14 on circulant/Cayley graphs); identifying what makes the icosahedron exact is the next target.

Every green number in this document stands because the gold ones are labelled gold.

OPEN

Four walls, and the files that carry them

WALL 0 — IS IT THE OPERATOR?

A plain Laplacian reproduces the coupling. Find what in the graph carries it, or the law belongs to the graph and not to T_p. Fifty to a hundred graph families would permit the multivariate analysis seventeen cannot support. **This wall now comes first.**

WALL 1 — PROVE THE FLIP

Elevate the clustering mechanism — κ flips near 95 triangles — from computed pattern to proven statement for T-class operators, and separate F_{coh} from F_{mi} while doing it.

WALL 2 — THE P SEAM

Justify or replace p's dual role. The 1.8σ fixed point at $p^* = 0.7577$ is the only lead in the file set.

WALL 3 — DYNAMICS

The failure is diagnosed as structural: nothing in T, G or I(i,j) is directed. A successor needs a **directed object** — not a better propagation run on a symmetric one.

AND THE MOST TRACTABLE TARGET OF ALL

$\Lambda \sim F = -1.0000$ at machine precision, every seed, every configuration, looks like an exact identity rather than a correlation. Proving it analytically — via Schur/equivariance on the icosahedral representation, the same route as Theorem 0 for $\kappa_0 = -0.0506$ — would convert a measured fact into a theorem.

COMPANION FILES · REPRODUCTION

AI33-MPOPT · Quantum-Gravity Bridge (the law, 36 pp) · **AI33-MPOPT · Loop Quantum Gravity Bridge** (the LQG comparison, 17 pp) · **AI33-MPOPT · Lawfulness** (the four-representation run, 28 pp) · this build-up document.

Seeds 0–4 throughout · exact W_1 via optimal transport, no entropic regularisation. All four files agree on: fifteen eliminated candidates, the operator not being the source, metric and t requiring declaration, the apex/curvilinear naming, the 1.7% band rarity, and the Void.

Scope, unchanged from first page to last: an exact operator on 32 and 33 nodes, computed end to end. Nobody was testing information–geometry recovery on a 32-node vertex–face incidence structure until AI33-MPOPT built one — and it produced a measurable, reproducible law.